

Assignment 2: Naive Bayes Classification

Course: CS4101 | CS6103

Deadline: 10th September, 2026

Instructions

1. Group members must be same as it was for Assignment 1.
2. Code must be properly indented, commented, and well-structured.
3. Marks will be deducted in case of plagiarism.
4. All required files must be compressed and named as:
 `rollno1_rollno2_rollno3.zip`
 e.g., `2021cs11_2021cs03_2021cs05.zip`.
5. Upload the assignment to: <https://forms.gle/hXJinTUMwqKfa1EN7>.

Problem Statement

You are required to build a clinical risk categorization system using two variants of the Naive Bayes classifier:

- Multinomial Naive Bayes
- Multivariate Bernoulli Naive Bayes

Download the UCI Heart Disease dataset from <https://archive.ics.uci.edu/dataset/45/heart+disease>. Use the Cleveland data, which contains 303 patient records and 14 commonly used attributes. The target attribute `num` represents the presence and severity of heart disease, with the following classes:

Value	Meaning
0	No presence of heart disease
1	Presence, severity 1
2	Presence, severity 2
3	Presence, severity 3
4	Presence, severity 4

The input attributes are: `age`, `sex`, `cp`, `trestbps`, `chol`, `fbs`, `restecg`, `thalach`, `exang`, `oldpeak`, `slope`, `ca`, and `thal`. The dataset uses “?” for missing values. You must document every cleaning decision and must not use the target attribute as an input feature.

Feature Representation

Naive Bayes models in this assignment must operate on discrete events. Keep the original categorical attributes as categorical events. Convert each continuous attribute (`age`, `trestbps`, `chol`, `thalach`, and `oldpeak`) into bins using thresholds learned from the training portion only. Use at least two alternative binning strategies:

1. equal-width binning with a clearly stated number of bins; and
2. quantile binning with a clearly stated number of bins.

The same training-derived thresholds must be applied unchanged to validation and test records. Represent a patient as a bag of feature-value tokens such as `age_bin=2` or `cp=4`. A missing value must be represented by an explicit token (for example, `thal=missing`) rather than silently discarded.

Task 1: Probability and Representation Analysis

1. Explain the Naive Bayes conditional-independence assumption for this dataset. Give two medically plausible examples where the assumption may be violated.
2. For one patient in the supplied data, show the complete transformation from raw attributes to discrete feature-value tokens under both binning strategies.
3. Select one record from the training data and calculate, by hand, the prior and at least three smoothed conditional probabilities for two target classes. State all counts and denominators.
4. Explain why a zero count can dominate the posterior decision and how Laplace smoothing changes that decision.

Task 2: Implement Naive Bayes from Scratch

Implement both models without calling a pre-built classifier:

1. Multinomial Naive Bayes, using the count of each feature-value token in each class.
2. Multivariate Bernoulli Naive Bayes, using the presence or absence of each vocabulary token in a patient record.

For each model, report the class prior, the vocabulary size, the smoothing formula, and the exact log-posterior score used for prediction. Include a deterministic tie-breaking rule for equal scores. For the Bernoulli model, account for both present and absent vocabulary events; do not evaluate only the tokens that appear in the test record.

Task 3: Experimental Comparison

Use the same folds and preprocessing protocol to compare the two models under both binning strategies. Investigate the following questions:

1. Does retaining the five original target severity classes produce a different conclusion from merging classes 1–4 into a single “disease present” class? Perform both experiments and explain the effect of class imbalance.
2. How do equal-width and quantile binning affect the learned likelihoods and predictions? Identify at least one patient for which the prediction changes and explain why.
3. Compare models trained with and without the explicit missing-value token. Discuss whether removing incomplete records is preferable for this dataset and support the discussion with results.
4. Compute and interpret the posterior scores for at least two validation patients, including one correctly classified and one misclassified patient.

Task 4: Priors, Smoothing, and Calibration

Use the validation data to study model behavior rather than selecting a model from the test set.

1. Compare empirical class priors with uniform class priors. Report the change in macro F1 and explain why accuracy alone may conceal this change.
2. Compare Laplace smoothing with at least one additional positive smoothing value, such as $\alpha = 0.5$ and $\alpha = 2$. Plot validation performance against α for both models.
3. Convert log scores into normalized posterior probabilities using a numerically stable procedure. Report multiclass log loss and a reliability diagram or another suitable calibration analysis.
4. Add a shuffled feature-value token column generated using a fixed random seed. Explain whether either model is sensitive to this irrelevant feature and why.

Task 5: Final Model and Clinical Error Analysis

Select one final preprocessing and model configuration using only the training and validation results. Retrain it on the permitted development data and evaluate it once on the held-out test portion. In addition to the metrics below, analyze errors:

1. Identify at least three false predictions and list their true class, predicted class, posterior probabilities, and relevant feature-value tokens.
2. For the binary experiment, discuss the relative cost of false negatives and false positives in a screening setting. Do not make medical claims beyond what is supported by the dataset.
3. State at least two limitations of using this dataset and Naive Bayes for clinical risk categorization.

Implementation Requirements

1. Text Preprocessing
 - Tokenization, lowercasing
 - Stopword removal
 - Vocabulary construction
2. Model Implementation
 - Implement both Multinomial and Multivariate Bernoulli Naive Bayes from scratch.
 - Use add-one (Laplace) smoothing.
 - Use log-probabilities to avoid numerical underflow.
3. Training & Testing
 - Perform 5-fold cross-validation.
 - Train the model on each fold and evaluate on the held-out test set.
4. Evaluation Metrics
 - Accuracy
 - Precision, Recall, F1-score
 - Confusion Matrix

- Macro and Micro F1-scores
- Plot Precision-Recall and ROC curves.

Expected Output

- Confusion matrix for each fold and final averaged results.
- Accuracy, Precision, Recall, and F1-score for both models.
- A comparison table for Multinomial vs. Multivariate Naive Bayes.
- Plot ROC curves.
- All the outputs must be clearly visible in the submitted notebook.

Restrictions

- No pre-built Naive Bayes libraries are allowed (e.g., scikit-learn).
- Only standard Python libraries (math, re, collections, matplotlib, numpy) are permitted.
- Tokenization and stopword removal can use libraries such as NLTK or spaCy, but classification must be implemented manually.

Reproducibility and Submission

- Random seeds must be explicitly set everywhere randomness is involved, including fold creation, binning when applicable, feature shuffling, and any sampling.
- The notebook must execute from beginning to end and reproduce all tables, scores, and figures reported in the submission.
- Submit one ZIP file containing exactly one implementation notebook and one PDF report, named using the roll numbers of all group members.